

Jiacheng Liang

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Research Interest

Stony Brook University Ph.D. Candidate specializing in **LLM Safety and Trustworthy AI**. Proven research impact with publications at top-tier security and AI conferences (**IEEE S&P[Security Top-1]**, EMNLP, ICLR, AsiaCCS). Internship experience from **Amazon AGI (Responsible AI)**, developing an adaptive red-teaming framework. Expertise in **LLM post-training, LLM Agent, Agent Memory, LLM Jailbreak/Backdoor Defense, LLM(MOE) Safety Alignment, RAG safety, LLM Watermark**.

Education

Stony Brook University & Penn State University

Ph.D. candidate; Department of Computer Science; GPA: 4.0/4.0

Stony Brook, NY

09/2022 – 12/2026(Expected)

University of Electronic Science and Technology of China

B.S. of Software Engineering - International Elite Class

Chengdu, China

09/2018 – 06/2022

Skills

Knowledge: LLM Agent, Large Language Model, LLM security, LLM Safety alignment, Agent Safety, GraphRAG, RAG, LoRA fine-tuning, LLM fine-tuning, post-training optimization, prompt engineering, inference optimization, LLM agents, Jailbreak defense, backdoor defense, Watermark in LLMs, Model distillation, Adversarial Machine Learning, Generative Models, Natural Language Processing

Languages: (Proficient) Python; (Familiar) C, SQL

Developer Tools: VS Code, Cursor, PyCharm, Conda, GitHub, Linux toolkits

Libraries / Frameworks: PyTorch, Huggingface, Transformers, OpenAI, TRL, VERL

Work Experience

Amazon AGI Foundation (Responsible AI Team)

05/2025 – 08/2025

Applied Scientist Intern

Boston, MA

- Developed an **adaptive red-teaming** framework to identify policy-reward failures in RLHF-trained LLMs.
- Designed a Red Mentor to expose systemic vulnerabilities and **enhance reward model** and policy robustness **via GRPO**.
- Improved safety alignment with minimal performance degradation.

Recent Academic Projects

Security Challenges in Large Language Models

10/2023 – Present

Research Assistant @ALPS-Lab (Dr. Ting Wang), Stony Brook University

Stony Brook, NY

- Potential Security Threats in GraphRAG:** Identify and analyze potential security threats within GraphRAG, focusing on vulnerabilities that could compromise the integrity of the entity relationship graphs and the overall knowledge base. (Pub. [1])
- Jailbreaking reasoning model:** Proposed an automated framework to jailbreak reasoning models by leveraging weak models to simulate reasoning processes and generate prompts. Achieved up to 100% jailbreak success rates on challenging LRM benchmarks. (Pub. [10])
- Evaluation Watermark's robustness and propose advanced attack method:** Investigate existing LLM watermark methods to form a Systematization of Knowledge(SoK) and establish a comprehensive evaluation platform to standardize their robustness. Propose an advanced method to attack the watermark and discuss the corresponding defenses. (Pub. [3])
- LLM Agent Security & Evaluation:** Developed AgentLAB, the first benchmark for evaluating LLM agents under long-horizon, adaptive attacks, covering 5 attack families across 28 realistic tool-using environments.
- LLM Agent Memory System Safety: Ongoing**

Defensive Strategies for Large Language Models

08/2024 – Present

Research Assistant @ALPS-Lab (Dr. Ting Wang), Stony Brook University

Stony Brook, NY

- Mixture of experts (MoE) LLM Safety Alignment:** Developed a dual-layered safety alignment framework for Mixture-of-Experts (MoE) models using a bi-level alternating optimization strategy to robustify experts and routers against adversarial jailbreaks.
- Backdoor Attack Defense for LLM Agents:** Developed a defense system that enables LLM agents to defend themselves against backdoor attacks by ensuring consistency between agent planning, execution, and user instructions. (Pub. [13])
- Jailbreak Defense Through Data Curation:** Created a defensive framework to mitigate jailbreaking attacks at every stage of LLM customization, achieving a 100% success rate in generating responsible responses. (Pub. [5])
- Jailbreak Attack Prevention Through KV Eviction:** Designed a defense against jailbreak attacks by selectively evicting

low-importance tokens from key-value caches, countering adversarial prompts while preserving LLM performance on benign queries. (Pub. [6])

- **Multimodal jailbreak defense:** Designed a defense mechanism against multimodal jailbreak attacks by reweighting visual tokens during inference. DTR is the first to apply KV cache optimization for robust multimodal LLM safety. (Pub. [11])

Model Extraction Security Challenges on the Real MLaaS API

08/2022 – 10/2023

Research Assistant @ALPS-Lab (Dr. Ting Wang), Stony Brook University

Stony Brook, NY

- **Model Extraction Platform Development:** Designed and launched "MEBench", an easy-to-use open-source evaluation tool, assessing ME vulnerabilities in various MLaaS APIs by integrating multiple attacks, metrics, and models. (Pub. [2])

Publications

- [1] **Liang, J.**, Wang, Y., Li, C., Zhu, R., Jiang, T., Gong, N., & Wang, T. (2025). GraphRAG under Fire. *IEEE Symposium on Security and Privacy 2026 (IEEE S&P 2026. Top-1 Security Conference)*. URL: <https://arxiv.org/abs/2501.14050>
- [2] **Liang, J.**, Pang, R., Li, C., & Wang, T. (2024). Model extraction attacks revisited. In *Proceedings of the 19th ACM ASIA CCS - Asia Conference on Computer and Communications Security* (pp. 1231-1245).
- [3] **Liang, J.**, Wang, Z., Hong, L., Ji, S., & Wang, T. (2025). WaterPark: A robustness assessment of language model watermarking. *EMNLP 2025*. url: <https://arxiv.org/abs/2411.13425>
- [4] **Liang, J.**, Li, S., Cao, B., Jiang, W., & He, C. (2021). Omnilytics: A blockchain-based secure data market for decentralized machine learning. *International Workshop on Federated Learning for User Privacy and Data Confidentiality in Conjunction with ICML 2021*
- [5] Liu, X. *, **Liang, J. ***, Tang, L., You, C., Ye, M., & Xi, Z. (2025). Data to Defense: The Role of Curation in Customizing LLMs Against Jailbreaking Attacks. *EMNLP 2025*. url: <https://arxiv.org/abs/2410.02220>
- [6] Jiang, T., Wang, Z., **Liang, J.**, Li, C., Wang, Y., & Wang, T. (2024). RobustKV: Defending large language models against jailbreak attacks via KV eviction. *The International Conference on Learning Representations (ICLR) 2025*. url: <https://arxiv.org/abs/2410.19937>
- [7] Zhou, Q., Guo, S., Pan, J., **Liang, J.**, Guo, J., Xu, Z., & Zhou, J. (2024). Pass: Patch automatic skip scheme for efficient on-device video perception. *IEEE TPAMI - IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- [8] Zhou, Q., Guo, S., Pan, J., **Liang, J.**, Xu, Z., & Zhou, J. (2023). PASS: patch automatic skip scheme for efficient real-time video perception on edge devices. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 37, No. 3, pp. 3787-3795)*.
- [9] Li, J., Wei, G., **Liang, J.**, Ren, Y., Lee, P. P., & Zhang, X. (2022). Revisiting frequency analysis against encrypted deduplication via statistical distribution. In *IEEE INFOCOM 2022 - IEEE Conference on Computer Communications* (pp. 290-299). *IEEE*.
- [10] Wang, Y., Li, C., Chen, G., **Liang, J.**, & Wang, T. (2025). Reasoning or Retrieval? A Study of Answer Attribution on Large Reasoning Models. *The International Conference on Learning Representations (ICLR) 2026*. url: <https://arxiv.org/abs/2509.24156>
- Works under review:
- [11] **Liang, J.**, Jiang, T., Wang, Y., Zhu, R., Ma, F., & Wang, T. (2025). AutoRAN: Automated Hijacking of Safety Reasoning in Large Reasoning Models. URL: <https://arxiv.org/abs/2505.10846>
- [12] Jiang, T., **Liang, J.**, Zhu, R., Zhou, J., Ma, F., & Wang, T. (2025). Robustifying Vision-Language Models via Dynamic Token Reweighting. URL: <https://arxiv.org/abs/2505.17132>
- [13] Zhu, R., Wang, Y., Jiang, T., **Liang, J.**, & Wang, T. (2025). Self-Improving Model Steering. URL: <https://arxiv.org/abs/2507.08967>
- [14] Li, C., **Liang, J.**, Cao, B., Chen, J., & Wang, T. (2025). Your Agent Can Defend Itself against Backdoor Attacks. URL: <https://arxiv.org/abs/2506.08336>
- [15] Xu, N., Li, C., Du, T., Li, M., Luo, W., **Liang, J.**, Li, Y., Zhang, X., Han, M., Yin, J., & Wang, T. (2024). CopyrightMeter: Revisiting copyright protection in text-to-image models. url: <https://arxiv.org/abs/2411.13144>